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The Editor, Icarus

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Dear Editor,

I submit the manuscript "Rotation periods and spin senses of asteroids from Gaia DR3 sparse photometry: calibrated reliability, null tests and disputed LCDB periods" for consideration as a regular paper in Icarus.

Gaia DR3 provides sparse photometry for about 157,000 asteroids, and several groups have derived periods from it. What is still missing is a measured error rate that transfers to asteroids not already in other catalogues. The manuscript provides one.

All choices are fixed on a tuning half of the known periods or by null tests that keep the real Gaia sampling. The reliability is then reported on an untouched test half: 89% of confident periods are correct, and 95% where the reference period is of high quality.

The phase-angle-bisector correction also yields a spin sense for each confident period. These senses agree with published poles for 94–98% of asteroids, and they reproduce the known Yarkovsky segregation in collisional families.

As by-products, the paper lists:

- 108 periods in the current LCDB that independent data or other Gaia analyses dispute (nine further cases have already been revised in the 2023 LCDB to the values found here);
- 861 candidate new periods, for observers to confirm.

The pipeline runs on all of DR3 in about 20 minutes and is ready for Gaia DR4.

The catalogue, code and validation tables are openly available on Zenodo (doi:10.5281/zenodo.23638281), where a preprint of the manuscript is also deposited. A short research note on a 600-asteroid pilot of an earlier version of the method was submitted to RNAAS; the present manuscript supersedes its method and validation.

Thank you for considering this work.

Sincerely,

Oleg Tropynin